=== src/m3/prepare_04_sec73.html === <!-- UNIT 3: BARTLE §7.3 (T20-T32) --> <section class="page"> UNIT 3 · BARTLE §7.3 <h1>The Fundamental Theorem</h1> <hr>
<!-- T20 (K20) --> T20<h2>Fundamental Theorem of Calculus (First Form)</h2> Evaluating \int_a^b f = F(b) - F(a) using the Mean Value Theorem (Bartle Theorem 7.3.1; PYQ MTS-26 Q8, MAT-22 MCQ 27, MAT-21 MCQ 19). Theorem 7.3.1 · Fundamental Theorem of Calculus, First Form (Bartle p. 216, eq. (1)) · say it word for wordSuppose there is a finite set E in [a,b] and functions f, F : [a,b] \to \mathbb{R} such that: (a) F is continuous on [a,b]; (b) F'(x) = f(x) for all x \in [a,b] \setminus E; (c) f belongs to \mathcal{R}[a,b]. Then we have
\int_a^b f = F(b) - F(a).
In plain words: if F is continuous on the whole closed interval [a,b], its derivative F' equals f except possibly at finitely many corner/endpoint exceptions E, and f is Riemann integrable, then the integral of f is simply F(b) - F(a). Hypotheses · Theorem 7.3.1 (each one is essential!) (a) F is continuous on all of [a,b] (even at points of E)let F(x) = \operatorname{sgn}(x) on [-1,1] and f(x) = 0; then F'(x) = 0 = f(x) on [-1,1]\setminus\{0\} (E=\{0\}) and f \in \mathcal{R}[-1,1], but F jumps at 0 and \int_{-1}^1 0 = 0 \ne 2 = F(1)-F(-1) (b) F'(x) = f(x) for all x \in [a,b] \setminus E where E is finiteif E is uncountable (e.g. the Cantor function F on [0,1], which is continuous with F'=0 off the Cantor set), \int_0^1 0 = 0 \ne 1 = F(1)-F(0) (c) f belongs to \mathcal{R}[a,b] (not automatic even when E = \varnothing!)K(x) = x^2\cos(1/x^2), K(0)=0 C19 is differentiable at every point of [0,1] (E=\varnothing), yet f = K' is unbounded near 0, so K' \notin \mathcal{R}[0,1]. Likewise H(x) = 2\sqrt x has H'(x) = 1/\sqrt x \notin \mathcal{R}[0,b] Skeleton · apply FTC 1 to evaluate \int_a^b fDefine $F(x) := $? on [a,b] and let $E := $? (finite). Check: (a) F is continuous on [a,b]; (b) F'(x) = f(x) for all x \in [a,b]\setminus E; (c) f \in \mathcal{R}[a,b] because ?. By FTC 1 (Theorem 7.3.1), $\int_a^b f = F(b) - F(a) = $?. ∎ </section>
<section class="page"> KnowF continuous on [a,b], F'(x) = f(x) on (a,b) (case E = \{a,b\}), and f \in \mathcal{R}[a,b].WantShow \big\lvert F(b) - F(a) - \int_a^b f\big\rvert < \varepsilon for every \varepsilon > 0 by rewriting F(b) - F(a) as a Riemann sum S(f;\dot P_u) via the Mean Value Theorem. Proof of Theorem 7.3.1 (Bartle pp. 216–217) We first prove the case where E = \{a,b\}. Let \varepsilon > 0 be given. Since f \in \mathcal{R}[a,b] by hypothesis (c), there exists \delta_\varepsilon > 0 such that if \dot P is any tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon, then
\left\lvert S(f;\dot P) - \int_a^b f\right\rvert < \varepsilon.
Definition 7.1.1 (T03) applied to f \in \mathcal{R}[a,b]. Let P = \{[x_{i-1},x_i]\}_{i=1}^n be a partition of [a,b] with \lVert P\rVert < \delta_\varepsilon. Since F is continuous on [x_{i-1},x_i] by (a) and differentiable on (x_{i-1},x_i) \subseteq [a,b]\setminus E by (b), the Mean Value Theorem 6.2.4 (P12) applied to F on [x_{i-1},x_i] provides a point u_i \in (x_{i-1},x_i) such that
F(x_i) - F(x_{i-1}) = F'(u_i)(x_i - x_{i-1}) = f(u_i)(x_i - x_{i-1}) \quad \text{for } i = 1, \dots, n.

Mean Value Theorem 6.2.4 on each [x_{i-1},x_i]; since u_i \in (x_{i-1},x_i) \subseteq (a,b), u_i \notin E, so F'(u_i) = f(u_i). Adding these n equations and telescoping the left-hand side gives
F(b) - F(a) = \sum_{i=1}^n \big(F(x_i) - F(x_{i-1})\big) = \sum_{i=1}^n f(u_i)(x_i - x_{i-1}) = S(f;\dot P_u),
where \dot P_u := \{([x_{i-1},x_i],u_i)\}_{i=1}^n is a tagged partition of [a,b] with norm \lVert P\rVert < \delta_\varepsilon.Telescoping sum (M01, M05) and definition of Riemann sum (T02). Substituting S(f;\dot P_u) = F(b) - F(a) into the inequality of Step 1 yields
\left\lvert F(b) - F(a) - \int_a^b f\right\rvert < \varepsilon.
Since \varepsilon > 0 is arbitrary, we conclude that \int_a^b f = F(b) - F(a).Arbitrary \varepsilon rule (P05). General finite set E (Exercise 7.3.1): Let a = c_0 < c_1 < \cdots < c_m = b be the points of E \cup \{a,b\} in increasing order. On each subinterval [c_{k-1},c_k], F is continuous, F'(x) = f(x) on (c_{k-1},c_k), and f|_{[c_{k-1},c_k]} \in \mathcal{R}[c_{k-1},c_k] by Corollary 7.2.10. Applying the case just proved on each [c_{k-1},c_k] and summing via Corollary 7.2.11 gives \int_a^b f = \sum_{k=1}^m \big(F(c_k) - F(c_{k-1})\big) = F(b) - F(a). ∎Corollaries 7.2.10 and 7.2.11 (T18) plus telescoping over the m pieces.   Exam trap · PYQ MAT-22 MCQ 27, MAT-21 MCQ 19 In the MCQ asking for \int_a^b f(x)\,dx when f is continuous and F is an antiderivative of f, the answer is simply F(b) - F(a)—do not multiply by (b-a)! In the model paper: Q14, Q29 (PYQ MTS-26 Q8, MAT-22 MCQ 27, MAT-21 MCQ 19). </section>
<!-- T21 (K21) --> <section class="page"> T21<h2>Examples and limits of FTC 1: when it works and when it fails</h2> Checking all three hypotheses (a), (b), (c) in practice (Bartle Examples 7.3.2(a)–(e), Exercises 7.3.2–7.3.4, 7.3.8; PYQ MAT-22 Q35, MAT-22 MCQ 28, 29, MAT-21 MCQ 20). Examples 7.3.2(a)–(e) (Bartle pp. 217–218) (a) \int_a^b x\,dx = \tfrac12(b^2 - a^2): F(x) := \tfrac12 x^2 has F'(x) = x for all x \in [a,b] (E = \varnothing), and f(x) = x is continuous so f \in \mathcal{R}[a,b] (7.2.7). More generally (Ex 7.3.2), H_n(x) = \frac{x^{n+1}}{n+1} gives \int_a^b x^n\,dx = \frac{b^{n+1}-a^{n+1}}{n+1} with E = \varnothing. (b) \int_a^b \frac{1}{x^2+1}\,dx = \operatorname{Arctan} b - \operatorname{Arctan} a: G(x) := \operatorname{Arctan} x has G'(x) = \frac{1}{x^2+1} on [a,b] (E = \varnothing), which is continuous and hence in \mathcal{R}[a,b]. (c) \int_{-10}^{10} \operatorname{sgn}(x)\,dx = 0: A(x) := \lvert x\rvert is continuous on [-10,10], A'(x) = \operatorname{sgn}(x) on [-10,10]\setminus\{0\} (E = \{0\}), and \operatorname{sgn} is a step function so \operatorname{sgn} \in \mathcal{R}[-10,10] (7.2.5). FTC 1 gives A(10) - A(-10) = 10 - 10 = 0. (d) Failure for H(x) := 2\sqrt x on [0,b]: H is continuous on [0,b] and H'(x) = 1/\sqrt x on (0,b) (E = \{0\}), but h = H' is unbounded on (0,b), so h \notin \mathcal{R}[0,b] (Theorem 7.1.6) no matter how h(0) is defined. Hypothesis (c) fails; FTC 1 does not apply. (e) Failure for K(x) := x^2\cos(1/x^2) (x \in (0,1]), K(0) := 0 C19: By the Product and Chain Rules, K'(x) = 2x\cos(1/x^2) + (2/x)\sin(1/x^2) for x > 0, and at 0, \lim_{x\to 0}\frac{K(x)-K(0)}{x} = \lim_{x\to 0} x\cos(1/x^2) = 0, so K'(0) = 0. Thus K is continuous and differentiable at every point of [0,1] (E = \varnothing)! However, K' is unbounded on [0,1] (at x_m = (\frac{\pi}{2} + 2m\pi)^{-1/2} \to 0, the second term is 2/x_m \to \infty), so K' \notin \mathcal{R}[0,1] by Theorem 7.1.6 and FTC 1 does

not apply. In plain words: having an antiderivative F is never enough by itself—you must always check that F is continuous on [a,b] and that f = F' is bounded and integrable! Worked examples · Three past-paper FTC 1 problems (Q14, Q15) 1. Bartle Ex 7.3.3 (PYQ MAT-22 MCQ 28, MAT-21 MCQ 20): Let g(x) = x for \lvert x\rvert \ge 1 and g(x) = -x for \lvert x\rvert < 1, and let G(x) = \tfrac12\lvert x^2 - 1\rvert on [-2,3]. Then G is continuous on [-2,3]; for \lvert x\rvert > 1, G(x) = \tfrac12(x^2-1) has G'(x) = x = g(x), and for \lvert x\rvert < 1, G(x) = \tfrac12(1-x^2) has G'(x) = -x = g(x), so G'(x) = g(x) off the finite set E = \{-1, 1\}. Also g is continuous on [-2,-1], [-1,1], and [1,3] (up to the jump at -1), so g \in \mathcal{R}[-2,3]. By FTC 1,
\int_{-2}^3 g(x)\,dx = G(3) - G(-2) = \frac12\lvert 9 - 1\rvert - \frac12\lvert 4 - 1\rvert = 4 - \frac32 = \frac52.
 2. Bartle Ex 7.3.4 (PYQ MAT-22 MCQ 29): Let B(x) = -\tfrac12 x^2 for x < 0 and B(x) = \tfrac12 x^2 for x \ge 0. For x < 0, B'(x) = -x = \lvert x\rvert; for x > 0, B'(x) = x = \lvert x\rvert; and at x = 0, both one-sided derivatives equal 0 = \lvert 0\rvert. Thus B'(x) = \lvert x\rvert for all x \in [a,b] (E = \varnothing), and \lvert x\rvert is continuous so \lvert x\rvert \in \mathcal{R}[a,b]. By FTC 1, \int_a^b \lvert x\rvert\,dx = B(b) - B(a). 3. PYQ MAT-22 Q35: Evaluate \int_0^1 x^2(1 - \sqrt x)\,dx. Write f(x) = x^2 - x^{5/2}, which is continuous on [0,1] and hence in \mathcal{R}[0,1] (Theorem 7.2.7). Define F(x) := \tfrac13 x^3 - \tfrac27 x^{7/2} on [0,1]; then F is continuous on [0,1] and F'(x) = x^2 - x^{5/2} = f(x) on [0,1] (E = \varnothing). By FTC 1, \int_0^1 x^2(1-\sqrt x)\,dx = F(1) - F(0) = \big(\tfrac13 - \tfrac27\big) - 0 = \dfrac{1}{21}. In the model paper: Q10, Q14, Q15 (PYQ MAT-22 Q35, MAT-22 MCQ 28, 29, MAT-21 MCQ 20). </section>
<!-- T22 (K22) --> <section class="page"> T22<h2>The indefinite integral is Lipschitz, hence continuous</h2> Accumulating area from a to z smooths out any integrable function f \in \mathcal{R}[a,b] into a Lipschitz function F(z) = \int_a^z f (Bartle Definition 7.3.3, Theorem 7.3.4; PYQ MAT-21 Q6). <figure> {{fig:F18_lipschitz_bound}} <figcaption>What to notice: For f(t)=t on [0,2] (M=2) between w=0.5 and z=1.5, the shaded area F(z)-F(w)=\int_w^z f = 1 sits inside the bounding box of height M=2 and width z-w=1 (area 2). What it does not prove: Being Lipschitz makes F continuous everywhere, but it does not guarantee F is differentiable where f jumps.</figcaption> </figure> Definition 7.3.3 & Theorem 7.3.4 (Bartle p. 218, eq. (3)) · say them word for word Definition 7.3.3: If f \in \mathcal{R}[a,b], the function defined by
F(z) := \int_a^z f \quad \text{for } z \in [a,b]
is called the indefinite integral of f with basepoint a. Theorem 7.3.4: The indefinite integral F defined by F(z) := \int_a^z f is continuous on [a,b]. In fact, if \lvert f(x)\rvert \le M for all x \in [a,b], then
\lvert F(z) - F(w)\rvert \le M\lvert z - w\rvert \quad \text{for all } z, w \in [a,b].
 In plain words: because f \in \mathcal{R}[a,b] is bounded by M, the area added between w and z is at most height M times width \lvert z - w\rvert, so F is Lipschitz (and therefore uniformly continuous) on [a,b]—even if f itself is discontinuous! Hypotheses · Theorem 7.3.4 f \in \mathcal{R}[a,b] (which by Theorem 7.1.6 automatically supplies M > 0 with \lvert f(x)\rvert \le M)no continuity of f is required here: even for Thomae C8 or \operatorname{sgn} C9, the indefinite integral F is continuous on [a,b] Proof of Theorem 7.3.4 (Bartle p. 218) <div

class="step">By Corollary 7.2.10, for each z \in [a,b] the restriction of f to [a,z] is in \mathcal{R}[a,z], so F(z) = \int_a^z f is well-defined (with F(a) = \int_a^a f = 0). By Theorem 7.1.6, f is bounded on [a,b], say \lvert f(x)\rvert \le M for all x \in [a,b].Corollary 7.2.10 (T18) and Boundedness Theorem 7.1.6 (T08). Assume w, z \in [a,b] with w \le z. By the Additivity Theorem 7.2.9,
F(z) = \int_a^z f = \int_a^w f + \int_w^z f = F(w) + \int_w^z f, \quad \text{so} \quad F(z) - F(w) = \int_w^z f.
Additivity Theorem 7.2.9 (T17) applied on [a,z] split at w. Since -M \le f(x) \le M for all x \in [w,z], Theorem 7.1.5(c) and Example 7.1.4(a) give
-M(z - w) \le \int_w^z f \le M(z - w),
whence \lvert F(z) - F(w)\rvert = \left\lvert\int_w^z f\right\rvert \le M\lvert z - w\rvert (and by symmetry the same holds if z < w).Monotonicity of the integral (7.1.5c) and constant integral (7.1.4a). Thus F satisfies a Lipschitz condition with constant M on [a,b], and hence F is continuous on [a,b] (given \varepsilon > 0, take \delta := \varepsilon/(M+1)). ∎Every Lipschitz function is continuous (P09).   Worked example · Changing the basepoint (Bartle Exercises 7.3.6, 7.3.9) If c \in [a,b] and F_c(z) := \int_c^z f is the indefinite integral with basepoint c, Theorem 7.2.13 gives F_a(z) = \int_a^z f = \int_a^c f + \int_c^z f = F_a(c) + F_c(z), so F_c(z) = F_a(z) - F_a(c): changing the basepoint simply shifts F by the constant -F_a(c). Similarly, \int_x^b f = F_a(b) - F_a(x) and \int_x^{\sin x} f = F_a(\sin x) - F_a(x). In the model paper: Q11 (PYQ MAT-21 Q6). </section>
<!-- T23 (K23) --> <section class="page"> T23<h2>Fundamental Theorem of Calculus (Second Form)</h2> Differentiating the area accumulator F(z) = \int_a^z f at a point c where f is continuous gives F'(c) = f(c) (Bartle Theorems 7.3.5–7.3.6; PYQ MTS-25 Q25, MAT-21 Q37a, MAT-22 Q33, MAT-22 MCQ 30, MTS-22 Q14). <figure> {{fig:F11_area_sliver}} <figcaption>What to notice: Over the thin slice [c,c+h] (c=1, h=0.2), continuity of f traps f(x) between f(c)-\varepsilon and f(c)+\varepsilon, so the sliver area F(c+h)-F(c)=\int_c^{c+h} f is trapped between (f(c)-\varepsilon)h and (f(c)+\varepsilon)h; dividing by h traps the difference quotient within \varepsilon of f(c). What it does not prove: If f jumps at c, the left and right slivers have different heights and F'(c) fails to exist.</figcaption> </figure> Theorem 7.3.5 & Theorem 7.3.6 · FTC Second Form (Bartle p. 219) · say them word for word Theorem 7.3.5 (Fundamental Theorem of Calculus, Second Form): Let f \in \mathcal{R}[a,b] and let f be continuous at a point c \in [a,b]. Then the indefinite integral F(z) := \int_a^z f is differentiable at c and F'(c) = f(c). Theorem 7.3.6: If f is continuous on [a,b], then the indefinite integral F(z) := \int_a^z f is differentiable on [a,b] and F'(x) = f(x) for all x \in [a,b]. In plain words: at every point c where an integrable function f is continuous, the rate of change of the accumulated area F(z) = \int_a^z f equals the height f(c) of the curve at c. If f is continuous on the whole interval [a,b], then F is an antiderivative of f on [a,b]. Hypotheses · Theorem 7.3.5 f \in \mathcal{R}[a,b]needed so that F(z) = \int_a^z f is defined on [a,b] f is continuous at the point cfor f = \operatorname{sgn} on [-1,1] C20, f is discontinuous at c=0 and F(x) = \lvert x\rvert - 1 is not differentiable at 0; for Thomae h C8, H(x) = \int_0^x h \equiv 0 is differentiable everywhere, but at every rational c \in \mathbb{Q}\cap[0,1], H'(c) = 0 \ne h(c)!

 Knowf \in \mathcal{R}[a,b], f continuous at c \in [a,b], and F(z) = \int_a^z f.WantGiven \varepsilon > 0, find \eta_\varepsilon > 0 so that 0 < \lvert h\rvert < \eta_\varepsilon \Rightarrow \left\lvert\dfrac{F(c+h) - F(c)}{h} - f(c)\right\rvert \le \varepsilon. Proof of Theorem 7.3.5 (Bartle pp. 219–220) Suppose first that c \in [a,b) and consider the right-hand derivative of F at c. Since f is continuous at c, given \varepsilon > 0 there exists \eta_\varepsilon > 0 such that if c \le x < c + \eta_\varepsilon (x \in [a,b]), then
f(c) - \varepsilon < f(x) < f(c) + \varepsilon.
Definition of continuity of f at c (Module 1, 5.1.1). Let h be any number with 0 < h < \eta_\varepsilon (and c + h \le b). By the Additivity Theorem 7.2.9 (and Corollary 7.2.10), f is integrable on [a,c], [a,c+h] and [c,c+h], and F(c+h) - F(c) = \int_a^{c+h} f - \int_a^c f = \int_c^{c+h} f.
Additivity Theorem 7.2.9 (T17). On the interval [c,c+h], the inequality of Step 1 holds, so Theorem 7.1.5(c) and Example 7.1.4(a) yield (f(c) - \varepsilon)\cdot h \le F(c+h) - F(c) = \int_c^{c+h} f \le (f(c) + \varepsilon)\cdot h.
Monotonicity of the integral (7.1.5c) with constant bounds f(c)\pm\varepsilon over an interval of length h. Dividing through by h > 0 and subtracting f(c) gives
\left\lvert\frac{F(c+h) - F(c)}{h} - f(c)\right\rvert \le \varepsilon.
Since \varepsilon > 0 is arbitrary, \lim_{h\to 0^+} \dfrac{F(c+h) - F(c)}{h} = f(c).h > 0 preserves inequalities; definition of right-hand limit. If c \in (a,b], the left-hand limit \lim_{h\to 0^-} \dfrac{F(c+h) - F(c)}{h} = f(c) is proved in the exact same way (for -\eta_\varepsilon < h < 0, on [c+h,c] we have (f(c)-\varepsilon)(-h) \le \int_{c+h}^c f = -(F(c+h)-F(c)) \le (f(c)+\varepsilon)(-h), and dividing by -h > 0 gives the same bound). Thus F is differentiable at c and F'(c) = f(c). ∎Agreement of left-hand and right-hand derivatives at interior points (and one-sided derivative at an endpoint a or b). Theorem 7.3.6 follows immediately by applying Theorem 7.3.5 at every x \in [a,b].
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<section class="page"> Variable-limit integrals (Leibniz Rule, Bartle Ex 7.3.10–11, 7.3.15–16; PYQ MTS-22 Q14, MAT-22 MCQ 30)
1. Differentiating \int_a^{v(x)} f(t)\,dt (Bartle Ex 7.3.10; PYQ MTS-22 Q14 "Leibniz rule"): If f is continuous on [a,b] and v : [c,d] \to [a,b] is differentiable, define F(u) := \int_a^u f(t)\,dt. By Theorem 7.3.6, F'(u) = f(u). Since G(x) := \int_a^{v(x)} f(t)\,dt = (F \circ v)(x), the Chain Rule 6.1.6 gives G'(x) = F'(v(x))\,v'(x) = f(v(x))\,v'(x). More generally, for two differentiable limits u(x), v(x),
\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)\,dt = \frac{d}{dx}\big(F(v(x)) - F(u(x))\big) = f(v(x))\,v'(x) - f(u(x))\,u'(x).
 2. Bartle Ex 7.3.11 (Model Paper Q19): <ul> (a) F(x) = \int_0^{x^2} (1 + t^3)^{-1}\,dt: here f(t) = (1+t^3)^{-1} is continuous on [0,1] and v(x) = x^2 has v'(x) = 2x, so F'(x) = f(x^2)\cdot 2x = \dfrac{2x}{1 + x^6}. (b) F(x) = \int_{x^2}^x \sqrt{1 + t^2}\,dt: here f(t) = \sqrt{1+t^2} is continuous, so F'(x) = f(x)\cdot 1 - f(x^2)\cdot 2x = \sqrt{1 + x^2} - 2x\sqrt{1 + x^4}. 3. Bartle Ex 7.3.16 (PYQ MAT-22 MCQ 30, Model Paper Q17): Suppose f : [0,1] \to \mathbb{R} is continuous and \int_0^x f = \int_x^1 f for all x \in [0,1]. Since f is continuous, f \in \mathcal{R}[0,1]. Let F(x) := \int_0^x f; by Additivity, \int_x^1 f = F(1) - F(x), so the hypothesis reads F(x) = F(1) - F(x), or 2F(x) = F(1) (a constant) for all x \in [0,1]. Differentiating via Theorem 7.3.6 gives 2f(x) = 2F'(x) = 0, so f(x) = 0 for all x \in [0,1].

In the model paper: Q17, Q19, Q40 (PYQ MTS-25 Q25, MAT-21 Q37a, MAT-22 Q33, MAT-22 MCQ 30, MTS-22 Q14). </section>
<!-- T24 (K24) --> <section class="page"> T24<h2>The indefinite integral need not be an antiderivative</h2> What happens to F(z) = \int_a^z f when f is not continuous: corners and ignored spikes (Bartle Examples 7.3.7(a),(b), Exercises 7.3.12–7.3.13). <figure> {{fig:F10_sgn_indefinite}} <figcaption>What to notice: The step function f(x)=\operatorname{sgn} x jumps from -1 to +1 at x=0; its indefinite integral F(x)=\lvert x\rvert (basepoint 0) or \lvert x\rvert-1 (basepoint -1) is continuous everywhere (Lipschitz with M=1) but has a sharp corner at x=0 where slope -1 meets slope +1. What it does not prove: Having F'(x) exist everywhere still does not make F an antiderivative of f—for Thomae's function, H'(x)=0 exists everywhere but disagrees with h(x) at every rational!</figcaption> </figure> Examples 7.3.7(a), (b) · When F is Not an Antiderivative (Bartle p. 219) (a) F'(c) fails to exist at a jump C20: Let f(x) := \operatorname{sgn} x on [-1,1]. Then f \in \mathcal{R}[-1,1] and its indefinite integral with basepoint -1 is F(x) = \int_{-1}^x \operatorname{sgn} = \lvert x\rvert - 1 for x \in [-1,1] (for x < 0, \int_{-1}^x (-1) = -(x+1) = \lvert x\rvert - 1; for x \ge 0, \int_{-1}^0 (-1) + \int_0^x 1 = -1 + x = \lvert x\rvert - 1). However, since F'(0) does not exist, F is not an antiderivative of f on [-1,1]. (b) F'(c) exists everywhere but does not equal f(c) C8: Let h be Thomae's function (Example 7.1.7) and let H(x) := \int_0^x h. Since \int_0^1 h = 0 and h \ge 0, we have H(x) = 0 for all x \in [0,1]. Thus H'(x) = 0 exists at every point of [0,1], but H'(x) = 0 \ne h(x) at every rational x \in \mathbb{Q}\cap[0,1], so H is not an antiderivative of h on [0,1]. In plain words: an antiderivative of f on [a,b] must satisfy F'(x) = f(x) at every point of [a,b]. If f jumps at c, F has a corner at c; if f has isolated spikes, F ignores them so F'(c) \ne f(c) at the spikes. Worked example · Bartle Exercise 7.3.12 Let f : [0,3] \to \mathbb{R} be f(x) = x on [0,1), f(x) = 1 on [1,2), and f(x) = x on [2,3]. Find F(x) = \int_0^x f and determine where F is differentiable: <ul> For 0 \le x \le 1: F(x) = \int_0^x t\,dt = \tfrac12 x^2. For 1 \le x \le 2: F(x) = \int_0^1 t\,dt + \int_1^x 1\,dt = \tfrac12 + (x - 1) = x - \tfrac12. For 2 \le x \le 3: F(x) = F(2) + \int_2^x t\,dt = \tfrac32 + \tfrac12(x^2 - 4) = \tfrac12(x^2 - 1). Notice that f is continuous at x = 1 (\lim_{x\to 1^-} x = 1 = f(1)), so by Theorem 7.3.5, F is differentiable at x = 1 with F'(1) = 1. At x = 2, f jumps from 1 to 2, so F'_-(2) = 1 \ne 2 = F'_+(2): F is differentiable on [0,3]\setminus\{2\} with F'(x) = f(x) there, and not differentiable at x = 2. In the model paper: Q12, Q40. </section>
<!-- T25 (K25) --> <section class="page"> T25<h2>The Substitution Theorem</h2> Reversing the Chain Rule to change variables in a definite integral (Bartle Theorem 7.3.8, Examples 7.3.9(a),(b), Exercises 7.3.17–7.3.19; PYQ MTS-25 Q6, MAT-22 Q18, MAT-21 Q19, MAT-22 MCQ 19, 26). <figure> {{fig:F12_substitution_panels}} <figcaption>What to notice: Setting x=\varphi(t)=\sqrt t transforms \int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt (left panel) into \int_1^2 2\sin x\,dx (right panel) with the exact same area 2(\cos 1-\cos 2) \approx 1.9129. What it does not prove: Theorem 7.3.8 cannot be used on [0,4] with \varphi(t)=\sqrt t because \varphi'(t)=1/(2\sqrt t) blows up at t=0; on [0,4] we must use FTC 1 with E=\{0\} instead.</figcaption> </figure> Theorem 7.3.8 · Substitution Theorem (Bartle p. 220, eq. (5)) · say it word for

wordLet J := [\alpha,\beta] and let \varphi : J \to \mathbb{R} have a continuous derivative on J. If f : I \to \mathbb{R} is continuous on an interval I containing \varphi(J), then \int_\alpha^\beta f(\varphi(t))\,\varphi'(t)\,dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx. In plain words: if x = \varphi(t) has a continuous derivative \varphi'(t) on the entire closed interval [\alpha,\beta] and f is continuous on the range of \varphi, you may replace f(\varphi(t))\varphi'(t)\,dt by f(x)\,dx and change the limits from t=\alpha,\beta to x=\varphi(\alpha),\varphi(\beta). Hypotheses · Theorem 7.3.8 \varphi is differentiable on J = [\alpha,\beta] and \varphi' is continuous on the closed interval [\alpha,\beta]for \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt with \varphi(t) = \sqrt t (Example 7.3.9(b) & Ex 7.3.19(a),(b)), \varphi'(t) = 1/(2\sqrt t) does not exist at t = 0; for \varphi(t) = \lvert t\rvert on [-1,1] (Ex 7.3.19(c)), \varphi'(0) does not exist; for \varphi(t) = \operatorname{Arcsin} t on [0,1] (Ex 7.3.19(d)), \varphi'(t) = 1/\sqrt{1-t^2} blows up at t = 1 f is continuous on an interval I \supseteq \varphi(J)used in Step 2 of the proof so that F(u) = \int_{\varphi(\alpha)}^u f(x)\,dx satisfies F'(u) = f(u) everywhere on I by Theorem 7.3.6 Proof of Theorem 7.3.8 (Bartle Exercise 7.3.17, p. 224) Since f is continuous on the interval I \supseteq \varphi(J), define the indefinite integral F(u) := \int_{\varphi(\alpha)}^u f(x)\,dx for u \in I (using Definition 7.2.12 if u < \varphi(\alpha)).Theorem 7.2.7 (T15) and Definition 7.3.3 (T22). By the Second Form of the Fundamental Theorem (Theorem 7.3.6), F is differentiable on I and F'(u) = f(u) for all u \in I.Theorem 7.3.6 (T23) since f is continuous on I. Define the composite function H(t) := F(\varphi(t)) for t \in J = [\alpha,\beta]. Since \varphi is differentiable on J and F is differentiable at \varphi(t) \in I, the Chain Rule 6.1.6 (P11) gives
H'(t) = F'(\varphi(t))\,\varphi'(t) = f(\varphi(t))\,\varphi'(t) \quad \text{for all } t \in J.
Chain Rule 6.1.6 (P11) and Step 2. Since f, \varphi, and \varphi' are continuous, the function t \mapsto f(\varphi(t))\varphi'(t) is continuous on J = [\alpha,\beta] and hence belongs to \mathcal{R}[\alpha,\beta] by Theorem 7.2.7.Composition and product of continuous functions are continuous; Theorem 7.2.7 (T15). Applying the First Form of the Fundamental Theorem (Theorem 7.3.1 with E = \varnothing) to H on [\alpha,\beta] and noting that H(\alpha) = F(\varphi(\alpha)) = \int_{\varphi(\alpha)}^{\varphi(\alpha)} f = 0 gives
\int_\alpha^\beta f(\varphi(t))\,\varphi'(t)\,dt = H(\beta) - H(\alpha) = F(\varphi(\beta)) - 0 = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx. \quad \blacksquare
Theorem 7.3.1 (T20) and Definition 7.2.12.
</section>
<section class="page"> Worked examples · Bartle Examples 7.3.9(a),(b) & Exercise 7.3.18 (PYQ MTS-25 Q6, MAT-22 Q18, MAT-21 Q19)
1. Example 7.3.9(a) (\int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt, PYQ MTS-25 Q6, MAT-21 Q19): On J = [1,4], set \varphi(t) := \sqrt t; then \varphi'(t) = \frac{1}{2\sqrt t} is continuous on [1,4]. With f(x) := 2\sin x (continuous on \mathbb{R}), we have f(\varphi(t))\varphi'(t) = (2\sin\sqrt t)\frac{1}{2\sqrt t} = \frac{\sin\sqrt t}{\sqrt t}, \varphi(1) = 1, and \varphi(4) = 2. By Theorem 7.3.8 and FTC 1,
\int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt = \int_1^2 2\sin x\,dx = \big[-2\cos x\big]_1^2 = 2(\cos 1 - \cos 2) \approx 1.912898.
 2. Example 7.3.9(b) (\int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt, PYQ MAT-22 Q18):

On [0,4], \varphi(t) = \sqrt t does not have a continuous derivative at t = 0, so Theorem 7.3.8 cannot be applied! Instead: (i) let g(t) := \frac{\sin\sqrt t}{\sqrt t} for t \in (0,4] and g(0) := 1; since \lvert\sin u\rvert \le \lvert u\rvert, \lvert g(t)\rvert \le 1 is bounded on [0,4] and continuous on every [c,4] (c>0), so g \in \mathcal{R}[0,4] by Exercise 7.2.11 (T18). (ii) Define F(t) := -2\cos\sqrt t on [0,4]; then F is continuous on [0,4] and F'(t) = (2\sin\sqrt t)\frac{1}{2\sqrt t} = g(t) for all t \in (0,4] (E = \{0\}). By FTC 1 (Theorem 7.3.1),
\int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt = F(4) - F(0) = -2\cos 2 - (-2\cos 0) = 2(1 - \cos 2) \approx 2.832294.
 3. Four drills from Bartle Ex 7.3.18: <ul> (a) \int_0^1 t\sqrt{1+t^2}\,dt: \varphi(t)=1+t^2, f(x)=\tfrac12\sqrt x \Rightarrow \int_1^2 \tfrac12 x^{1/2}\,dx = \big[\tfrac13 x^{3/2}\big]_1^2 = \dfrac{2\sqrt 2 - 1}{3} \approx 0.609476. (b) \int_0^2 t^2(1+t^3)^{-1/2}\,dt: \varphi(t)=1+t^3, f(x)=\tfrac13 x^{-1/2} \Rightarrow \int_1^9 \tfrac13 x^{-1/2}\,dx = \big[\tfrac23\sqrt x\big]_1^9 = \dfrac{4}{3}. (c) \int_1^4 \frac{\sqrt{1+\sqrt t}}{\sqrt t}\,dt: \varphi(t)=1+\sqrt t on [1,4], f(x)=2\sqrt x \Rightarrow \int_2^3 2x^{1/2}\,dx = \big[\tfrac43 x^{3/2}\big]_2^3 = \dfrac43(3\sqrt 3 - 2\sqrt 2) \approx 3.156967. (d) \int_1^4 \frac{\cos\sqrt t}{\sqrt t}\,dt: \varphi(t)=\sqrt t on [1,4], f(x)=2\cos x \Rightarrow \int_1^2 2\cos x\,dx = 2(\sin 2 - \sin 1) \approx 0.135653. 4. Symmetry and Translation (PYQ MAT-22 MCQ 19, 26; Q16): With \varphi(x) = \tfrac{\pi}{2} - x (\varphi' = -1), \int_0^{\pi/2} f(\cos x)\,dx = -\int_{\pi/2}^0 f(\sin u)\,du = \int_0^{\pi/2} f(\sin u)\,du. Splitting \int_0^\pi f(\sin x)\,dx = \int_0^{\pi/2} + \int_{\pi/2}^\pi and substituting u = \pi - x on [\pi/2,\pi] shows \int_{\pi/2}^\pi f(\sin x)\,dx = \int_0^{\pi/2} f(\sin u)\,du, so \int_0^{\pi/2} f(\cos x)\,dx = \tfrac12\int_0^\pi f(\sin x)\,dx (and is NOT equal to \int_0^\pi f(\sin x)\,dx). For the c-translate g(y) := f(y-c) on [a+c,b+c] (Bartle Ex 7.1.15), shifting any tagged partition of [a+c,b+c] by -c gives \int_{a+c}^{b+c} g = \int_a^b f. In the model paper: Q16, Q30 (PYQ MTS-25 Q6, MAT-22 Q18, MAT-21 Q19, MAT-22 MCQ 19, 26). </section>
<!-- T26 (K26) --> <section class="page"> T26<h2>Null sets and "almost everywhere"</h2> When is a set of bad points small enough for the Riemann integral to ignore? (Bartle Definition 7.3.10, Example 7.3.11; PYQ MTS-25 Q7, MTS-26 Q7, MTS-22 MCQ 10). <figure> {{fig:F17_rationals_cover}} <figcaption>What to notice: Enumerating \mathbb{Q}\cap[0,1]=\{r_1,r_2,r_3,\dots\} and covering r_k by an open interval J_k of width \varepsilon/2^k makes the total width \sum_{k=1}^\infty \varepsilon/2^k = \varepsilon, even though \mathbb{Q}\cap[0,1] is dense in [0,1]! What it does not prove: Being a null set does not mean the set is empty (\varnothing) or even countable—the Cantor set is uncountable and null.</figcaption> </figure> Definition 7.3.10 · Null Set and Almost Everywhere (Bartle p. 220, eq. (6)) · say it word for word (a) A set Z \subseteq \mathbb{R} is said to be a null set if for every \varepsilon > 0 there exists a countable collection \{(a_k,b_k)\}_{k=1}^\infty of open intervals such that
Z \subseteq \bigcup_{k=1}^\infty (a_k,b_k) \qquad \text{and} \qquad \sum_{k=1}^\infty (b_k - a_k) \le \varepsilon.
 (b) If Q(x) is a statement about the point x \in I, we say that Q(x) holds almost everywhere on I (or for almost every x \in I), written Q(x) for a.e. x \in I, if there exists a null set Z \subseteq I such that Q(x) holds for all x \in I \setminus Z. In plain words: a set Z is a null set (has "length zero") if, whatever tiny total length \varepsilon > 0 you are handed, you can cover all of Z with a sequence of open intervals whose lengths add up to at most \varepsilon. Proof of Example 7.3.11 · \mathbb{Q}\cap[0,1] is a null set (Bartle p. 221; PYQ MTS-25 Q7) <div

class="steps"> The set \mathbb{Q}_1 := \mathbb{Q} \cap [0,1] is countable (P13), so its elements can be enumerated as a sequence r_1, r_2, \dots, r_k, \dots.Countability of the rationals (P13). Let \varepsilon > 0 be given. For each k \in \mathbb{N}, let J_k be the open interval
J_k := \left(r_k - \frac{\varepsilon}{2^{k+1}},\, r_k + \frac{\varepsilon}{2^{k+1}}\right).
Then J_k contains r_k and has length b_k - a_k = 2\cdot\dfrac{\varepsilon}{2^{k+1}} = \dfrac{\varepsilon}{2^k}.The \varepsilon/2^k geometric covering trick (M07). The union \bigcup_{k=1}^\infty J_k of this countable collection of open intervals contains every point of \mathbb{Q}_1, and the sum of the lengths is \sum_{k=1}^\infty (b_k - a_k) = \sum_{k=1}^\infty \frac{\varepsilon}{2^k} = \varepsilon\sum_{k=1}^\infty \frac{1}{2^k} = \varepsilon\cdot 1 = \varepsilon.
Since \varepsilon > 0 is arbitrary, \mathbb{Q}_1 = \mathbb{Q} \cap [0,1] is a null set. ∎Geometric series \sum_{k=1}^\infty 1/2^k = 1 (P03) and Definition 7.3.10(a).   Result · Four key properties of null sets (Bartle p. 221; PYQ MTS-22 MCQ 10) 1. Subsets: Every subset A \subseteq Z of a null set Z is a null set (any open cover of Z also covers A). 2. Unions: The union Z_1 \cup Z_2 of two null sets (or countably many null sets) is a null set (cover Z_1 with total length \le \varepsilon/2 and Z_2 with total length \le \varepsilon/2). 3. Every countable set is null: The exact same \varepsilon/2^k argument of Example 7.3.11 shows that every finite or countably infinite set in \mathbb{R} is a null set. 4. Not every null set is countable (MTS-22 MCQ 10 trap!): The Cantor set (Bartle Definition 11.1.10) is an uncountable null set in [0,1]. On the other hand, a non-degenerate interval [a,b] (a < b) is never a null set (its length is b - a > 0). In the model paper: Q13 (PYQ MTS-25 Q7, MTS-26 Q7, MTS-22 MCQ 10). </section>
<!-- T27 (K27) --> <section class="page"> T27<h2>Lebesgue's Integrability Criterion</h2> The complete "if and only if" verdict on which bounded functions are Riemann integrable (Bartle Theorem 7.3.12, Examples 7.3.13(a)–(e); PYQ MAT-22 Q6). Theorem 7.3.12 · Lebesgue's Integrability Criterion (Bartle p. 221) · say it word for wordA bounded function f : [a,b] \to \mathbb{R} is Riemann integrable if and only if it is continuous almost everywhere on [a,b]. In plain words: f \in \mathcal{R}[a,b] if and only if (1) f is bounded on [a,b], and (2) the set D(f) = \{x \in [a,b] : f \text{ is discontinuous at } x\} is a null set. (The proof of 7.3.12 is in Bartle Appendix C and is not in the syllabus; we quote 7.3.12 to test functions and prove Theorems 7.3.14–7.3.16.) Hypotheses · Theorem 7.3.12 f is bounded on [a,b]f(x) = 1/x on (0,1], f(0) = 0 C16 has discontinuity set D(f) = \{0\} (a 1-point set, hence null!), yet f \notin \mathcal{R}[0,1] because f is unbounded! Never forget to check boundedness first. the discontinuity set D(f) is a null setDirichlet's function C7 is bounded (0 \le f \le 1), but D(f) = [0,1], which is not a null set, so f \notin \mathcal{R}[0,1] Skeleton · test integrability using Lebesgue's CriterionStep 1 (Boundedness): Check that \lvert f(x)\rvert \le M on [a,b] (if unbounded, f \notin \mathcal{R}[a,b] by Theorem 7.1.6). Step 2 (Discontinuity set): Find the set D of points of [a,b] where f is discontinuous: D = $?. Since D is finite / countable / null / not null, by Lebesgue's Integrability Criterion

(7.3.12), f is in / not in $R[a,b]$. ∎ Worked examples · Bartle Examples 7.3.13(a)–(e) (Model Paper Q31) (a) Step functions: The step function g of Example 7.1.4(b) is bounded (\lvert g\rvert \le 3$) and discontinuous only at x = 1. In general, any step function is bounded and has at most finitely many discontinuities (a null set), so every step function is in \mathcal{R}[a,b]. (b) Monotone functions: A monotone function on [a,b] is bounded between f(a) and f(b), and by Theorem 5.6.4 (P10) its set of discontinuities is at most countable, hence a null set. By 7.3.12, every monotone function on [a,b] is in \mathcal{R}[a,b]. (c) G(1/n) = 1/n (n \in \mathbb{N}), G(x) = 0 elsewhere on [0,1] (Example 7.1.4(d)): G is bounded (0 \le G \le 1), continuous at 0 (since \lim_{x\to 0} G(x) = 0 = G(0)), and discontinuous precisely on the countable set D = \{1, \tfrac12, \tfrac13, \dots, \tfrac1n, \dots\}. Since every countable set is a null set, G \in \mathcal{R}[0,1]. (d) Dirichlet's function C7: Discontinuous at every point of [0,1] (D = [0,1], not a null set), so f \notin \mathcal{R}[0,1]. (e) Thomae's function h C8: Bounded (0 \le h \le 1) and discontinuous precisely on D = \mathbb{Q} \cap [0,1], which is a null set by Example 7.3.11. By 7.3.12, h \in \mathcal{R}[0,1]. In the model paper: Q31 (PYQ MAT-22 Q6). </section>
<!-- T28 (K28) --> <section class="page"> T28<h2>The Composition Theorem</h2> Composing f \in \mathcal{R}[a,b] with an outer continuous function \varphi preserves integrability (Bartle Theorem 7.3.14; PYQ MAT-21 Q4). Theorem 7.3.14 · Composition Theorem (Bartle p. 222) · say it word for wordLet f \in \mathcal{R}[a,b] with f([a,b]) \subseteq [c,d], and let \varphi : [c,d] \to \mathbb{R} be continuous. Then the composition \varphi \circ f belongs to \mathcal{R}[a,b]. In plain words: if f is integrable and \varphi is continuous on the range of f, then \varphi(f(x)) is also integrable—because \varphi \circ f can only be discontinuous where f was already discontinuous! Hypotheses · Theorem 7.3.14 inner function f \in \mathcal{R}[a,b] and outer function \varphi : [c,d] \to \mathbb{R} is continuousWhy continuity of \varphi cannot be dropped (Bartle Remark p. 222 / Ex 7.3.22): Let h \in \mathcal{R}[0,1] be Thomae's function C8, and let \varphi : [0,1] \to \mathbb{R} be \varphi(0) := 0 and \varphi(y) := 1 for y \in (0,1]. Notice that \varphi \in \mathcal{R}[0,1] (discontinuous only at y=0, a single point!). Yet for x \in [0,1], (\varphi \circ h)(x) = 1 if x \in \mathbb{Q}\cap[0,1] (since h(x) = 1/n > 0) and (\varphi \circ h)(x) = \varphi(0) = 0 if x is irrational: \varphi \circ h is Dirichlet's function C7, which is NOT in \mathcal{R}[0,1]! Thus the composition of two integrable functions need not be integrable unless the outer one is continuous. Knowf \in \mathcal{R}[a,b] (so f is bounded and D(f) is null by 7.3.12), and \varphi : [c,d] \to \mathbb{R} is continuous.WantShow \varphi \circ f is bounded and D(\varphi \circ f) \subseteq D(f) is null, then invoke Lebesgue's Criterion 7.3.12. Proof of Theorem 7.3.14 (Bartle p. 222) If f is continuous at a point u \in [a,b], then since \varphi is continuous at f(u) \in [c,d], the composition \varphi \circ f is also continuous at u.Composition of continuous functions is continuous (Module 1, Theorem 5.2.6 / P07). Taking the contrapositive: if \varphi \circ f is discontinuous at u, then f must be discontinuous at u. Thus the set D of points of discontinuity of

\varphi \circ f on [a,b] is a subset of the set D(f) of points of discontinuity of f.u \notin D(f) \Rightarrow u \notin D, so D \subseteq D(f). Since f \in \mathcal{R}[a,b], Lebesgue's Integrability Criterion 7.3.12 implies that D(f) is a null set. Since any subset of a null set is a null set (T26), D is also a null set.Theorem 7.3.12 (\Rightarrow) and subset property of null sets (T26). Furthermore, since \varphi is continuous on the closed bounded interval [c,d], \varphi is bounded on [c,d] by the Boundedness Theorem 5.3.2 (P07), so \varphi \circ f is bounded on [a,b]. By Lebesgue's Integrability Criterion 7.3.12, \varphi \circ f \in \mathcal{R}[a,b]. ▮Boundedness Theorem 5.3.2 (P07) and Theorem 7.3.12 (\Leftarrow).   In the model paper: Q32 (PYQ MAT-21 Q4). </section>
<!-- T29 (K29) & T30 (K30) --> <section class="page"> T29<h2>Absolute value and the triangle inequality for integrals</h2> If f \in \mathcal{R}[a,b], then \lvert f\rvert \in \mathcal{R}[a,b] and \big\lvert\int_a^b f\big\rvert \le \int_a^b \lvert f\rvert \le M(b-a) (Bartle Corollary 7.3.15; PYQ MTS-25 Q8, MTS-22 Q7, MAT-21 Q20). Corollary 7.3.15 (Bartle p. 222) · say it word for wordSuppose f \in \mathcal{R}[a,b]. Then its absolute value \lvert f\rvert is in \mathcal{R}[a,b], and
$$\left\lvert\int_a^b f\right\rvert \le \int_a^b \lvert f\rvert \le M(b-a),$$
where \lvert f(x)\rvert \le M for all x \in [a,b]. In plain words: taking the absolute value of an integrable function keeps it integrable, and the absolute value of the integral never exceeds the integral of the absolute value (because positive and negative areas can cancel in \int_a^b f, but never cancel in \int_a^b \lvert f\rvert). Direction & Converse · Corollary 7.3.15 f \in \mathcal{R}[a,b] \Rightarrow \lvert f\rvert \in \mathcal{R}[a,b]Converse is FALSE: \lvert f\rvert \in \mathcal{R}[a,b] does not imply f \in \mathcal{R}[a,b]! Define f(x) := 1 for rational x \in [0,1] and f(x) := -1 for irrational x \in [0,1]. Then \lvert f(x)\rvert = 1 for all x \in [0,1], so \lvert f\rvert \in \mathcal{R}[0,1] with \int_0^1 \lvert f\rvert = 1, yet f \notin \mathcal{R}[0,1] (rational tags give S = 1, irrational tags give S = -1, gap = 2). Proof of Corollary 7.3.15 (Bartle p. 222) Since f \in \mathcal{R}[a,b], Theorem 7.1.6 implies that f is bounded on [a,b], say \lvert f(x)\rvert \le M for all x \in [a,b], so f([a,b]) \subseteq [-M,M].Boundedness Theorem 7.1.6 (T08). The function \varphi(t) := \lvert t\rvert is continuous on [-M,M]. By the Composition Theorem 7.3.14, \lvert f\rvert = \varphi \circ f belongs to \mathcal{R}[a,b].Continuity of \lvert t\rvert (Module 1, T04) and Composition Theorem 7.3.14 (T28). Since -\lvert f(x)\rvert \le f(x) \le \lvert f(x)\rvert for all x \in [a,b], Theorem 7.1.5(a) and (c) give
$$-\int_a^b \lvert f\rvert \le \int_a^b f \le \int_a^b \lvert f\rvert, \quad \text{which means} \quad \left\lvert\int_a^b f\right\rvert \le \int_a^b \lvert f\rvert.$$
Monotonicity and linearity of the integral (Theorem 7.1.5, T07) and P05. Finally, since \lvert f(x)\rvert \le M on [a,b], Theorem 7.1.5(c) and Example 7.1.4(a) yield \int_a^b \lvert f\rvert \le \int_a^b M = M(b-a). ▮Constant integral \int_a^b M = M(b-a) (T05).   In the model paper: Q33 (PYQ MTS-25 Q8, MTS-22 Q7, MAT-21 Q20).
<!-- T30 (K30) --> T30<h2>The Product Theorem</h2> The product fg of two integrable functions is integrable

via the polarisation identity fg = \tfrac12[(f+g)^2 - f^2 - g^2] (Bartle Theorem 7.3.16; PYQ MTS-26 Q6, MAT-22 Q17, MAT-21 Q18). Theorem 7.3.16 · Product Theorem (Bartle p. 222) · say it word for wordIf f and g belong to \mathcal{R}[a,b], then the product fg belongs to \mathcal{R}[a,b]. In plain words: squaring an integrable function keeps it integrable (by the Composition Theorem with \varphi(t) = t^2), and any product fg can be written in terms of squares as \tfrac12[(f+g)^2 - f^2 - g^2]. Hypotheses & Trap · Theorem 7.3.16 f, g \in \mathcal{R}[a,b] \Rightarrow fg \in \mathcal{R}[a,b]Converse is FALSE: f^2 \in \mathcal{R}[a,b] does NOT imply f \in \mathcal{R}[a,b] (take f = \pm 1 on rationals/irrationals: f^2 \equiv 1 \in \mathcal{R}, f \notin \mathcal{R}). Warning: \int_a^b fg \ne \left(\int_a^b f\right)\left(\int_a^b g\right) in general! For f(x) = g(x) = x on [0,1], \int_0^1 x^2\,dx = \tfrac13 \ne \tfrac12\cdot\tfrac12 = \tfrac14. Proof of Theorem 7.3.16 (Bartle p. 222) Since f, g \in \mathcal{R}[a,b], we also have f + g \in \mathcal{R}[a,b] by Theorem 7.1.5(b). By the Boundedness Theorem 7.1.6, f, g, and f+g are bounded on [a,b]; choose M > 0 so that f([a,b]), g([a,b]), and (f+g)([a,b]) all lie in [-M,M].Linearity 7.1.5(b) (T07) and Boundedness Theorem 7.1.6 (T08). The squaring function \varphi(t) := t^2 is continuous on [-M,M]. Applying the Composition Theorem 7.3.14 to f, g, and f+g shows that f^2 = \varphi \circ f, g^2 = \varphi \circ g, and (f+g)^2 = \varphi \circ (f+g) all belong to \mathcal{R}[a,b].Composition Theorem 7.3.14 (T28) with \varphi(t) = t^2. From the algebraic identity
fg = \frac12\big[(f + g)^2 - f^2 - g^2\big],
linearity of the integral (Theorem 7.1.5(a),(b)) implies that fg \in \mathcal{R}[a,b]. ∎Expand (f+g)^2 = f^2 + 2fg + g^2 and apply Theorem 7.1.5 (T07).   In the model paper: Q34 (PYQ MTS-26 Q6, MAT-22 Q17, MAT-21 Q18). </section>
<!-- T31 (K31) & T32 (K32) --> <section class="page"> T31<h2>Integration by Parts</h2> Integrating the Product Rule (FG)' = fG + Fg via the Product Theorem and FTC 1 (Bartle Theorem 7.3.17). Theorem 7.3.17 · Integration by Parts (Bartle p. 222, eq. (7)) · say it word for wordLet F, G be differentiable on [a,b] and let f := F' and g := G' belong to \mathcal{R}[a,b]. Then
\int_a^b fG = \Big[FG\Big]_a^b - \int_a^b Fg.
In plain words: if F and G are differentiable on [a,b] and their derivatives f = F' and g = G' are Riemann integrable on [a,b], then \int_a^b F'(x)G(x)\,dx = F(b)G(b) - F(a)G(a) - \int_a^b F(x)G'(x)\,dx. Hypotheses · Theorem 7.3.17 F, G are differentiable on [a,b] and f = F', g = G' \in \mathcal{R}[a,b]remember from Example 7.3.2(e) C19 that F differentiable does not automatically make F' \in \mathcal{R}[a,b], so f, g \in \mathcal{R}[a,b] must be assumed (it is automatic when f, g are continuous) Proof of Theorem 7.3.17 (Bartle p. 223) By the Product Rule for derivatives (Theorem 6.1.3(c), P11), the product FG is differentiable on [a,b] and
(FG)'(x) = F'(x)G(x) + F(x)G'(x) = f(x)G(x) + F(x)g(x) \quad \text{for all } x \in [a,b].
Product Rule 6.1.3(c) (P11). Since F and G are differentiable on [a,b], they are continuous on [a,b] by Theorem 6.1.2 (P11) and hence belong to \mathcal{R}[a,b] by Theorem 7.2.7. Since f, g \in \mathcal{R}[a,b] by hypothesis, the

Product Theorem 7.3.16 implies that fG and Fg both belong to \mathcal{R}[a,b].Theorem 6.1.2 (differentiable \Rightarrow continuous), Theorem 7.2.7 (T15), and Product Theorem 7.3.16 (T30). By Theorem 7.1.5(b), (FG)' = fG + Fg \in \mathcal{R}[a,b]. Applying the First Form of the Fundamental Theorem 7.3.1 (with E = \varnothing) to FG gives
\Big[FG\Big]_a^b = F(b)G(b) - F(a)G(a) = \int_a^b (FG)' = \int_a^b fG + \int_a^b Fg.
Subtracting \int_a^b Fg yields \int_a^b fG = \big[FG\big]_a^b - \int_a^b Fg. ∎Linearity 7.1.5(b) (T07) and FTC 1 (Theorem 7.3.1, T20).   Worked example · Evaluate \int_0^1 x e^x\,dx (Model Paper Q35) Take F(x) := e^x and G(x) := x on [0,1]. Both are differentiable on [0,1] with f(x) = F'(x) = e^x and g(x) = G'(x) = 1, which are continuous and hence in \mathcal{R}[0,1]. By Theorem 7.3.17 and FTC 1,
\int_0^1 e^x\cdot x\,dx = \big[e^x\cdot x\big]_0^1 - \int_0^1 e^x\cdot 1\,dx = (e\cdot 1 - 1\cdot 0) - \big[e^x\big]_0^1 = e - (e - 1) = 1.
  In the model paper: Q35.
</section>
<section class="page"> T32<h2>Taylor's Theorem with the Integral Remainder</h2> Repeated Integration by Parts yields an exact integral formula for the Taylor remainder R_n (Bartle Theorem 7.3.18; PYQ MTS-25 Q19, MAT-22 Q32). <figure> {{fig:F14_taylor_remainder}} <figcaption>What to notice: For f(x)=e^x on [0,1] (a=0, b=1), the linear Taylor polynomial T_1(1)=2 leaves remainder R_1 = \int_0^1 e^t(1-t)\,dt = e - 2 \approx 0.7183, and the quadratic polynomial T_2(1)=2.5 leaves remainder R_2 = \frac12\int_0^1 e^t(1-t)^2\,dt = e - 2.5 \approx 0.2183. What it does not prove: Unlike the Lagrange remainder (Module 2 / §6.4, out of scope), this formula has no unknown point c; it requires f^{(n+1)} \in \mathcal{R}[a,b].</figcaption> </figure> Theorem 7.3.18 · Taylor's Theorem with the Remainder (Bartle p. 223, eqs. (8)–(9)) · say it word for wordSuppose that f', \dots, f^{(n)}, f^{(n+1)} exist on [a,b] and that f^{(n+1)} \in \mathcal{R}[a,b]. Then we have
f(b) = f(a) + \frac{f'(a)}{1!}(b - a) + \cdots + \frac{f^{(n)}(a)}{n!}(b - a)^n + R_n,
where the remainder is given by
R_n = \frac{1}{n!}\int_a^b f^{(n+1)}(t)\,(b - t)^n\,dt.
In plain words: if you integrate R_n = \frac{1}{n!}\int_a^b f^{(n+1)}(t)(b-t)^n\,dt by parts n times, each step peels off one Taylor term -\frac{f^{(k)}(a)}{k!}(b-a)^k until only R_0 = \int_a^b f'(t)\,dt = f(b) - f(a) remains. Hypotheses · Theorem 7.3.18 f', f'', \dots, f^{(n+1)} exist at every point of [a,b] and f^{(n+1)} \in \mathcal{R}[a,b]for k \le n, f^{(k)} is differentiable on [a,b] (since f^{(k+1)} exists), hence continuous on [a,b] (Theorem 6.1.2) and in \mathcal{R}[a,b] (Theorem 7.2.7); only the top derivative f^{(n+1)} needs the extra assumption f^{(n+1)} \in \mathcal{R}[a,b] Proof of Theorem 7.3.18 (Bartle p. 223, with misprint corrected) Since f^{(n)} and t \mapsto (b-t)^n/n! are differentiable on [a,b] with derivatives f^{(n+1)} \in \mathcal{R}[a,b] and -(b-t)^{n-1}/(n-1)! \in \mathcal{R}[a,b] (continuous), we apply Integration by Parts (Theorem 7.3.17) to R_n with F(t) := f^{(n)}(t) (f(t) = F'(t) = f^{(n+1)}(t)) and G(t) := (b-t)^n/n! (g(t) = G'(t) = -(b-t)^{n-1}/(n-1)!).Check hypotheses of Theorem 7.3.17 (T31) for F(t) = f^{(n)}(t) and G(t) = (b-t)^n/n!. Since G(b) = (b-b)^n/n! = 0 and G(a) = (b-a)^n/n!, Theorem 7.3.17 gives
R_n = \left[\frac{f^{(n)}(t)(b-t)^n}{n!}\right]_{t=a}^{t=b} - \int_a^b

f^{(n)}(t)\left(-\frac{(b-t)^{n-1}}{(n-1)!}\right)dt = -\frac{f^{(n)}(a)}{n!}(b-a)^n + \frac{1}{(n-1)!}\int_a^b f^{(n)}(t)(b-t)^{n-1}\,dt.
That is, R_n = -\dfrac{f^{(n)}(a)}{n!}(b-a)^n + R_{n-1}, where R_{n-1} = \dfrac{1}{(n-1)!}\int_a^b f^{(n)}(t)(b-t)^{n-1}\,dt.Evaluate the boundary term at t=b and t=a. (Bartle misprint note: the first displayed integral on p. 223 prints (b-a)^{n-1} by mistake; the variable of integration is t, so it must be (b-t)^{n-1}, as Bartle's next line confirms.) For each k = n, n-1, \dots, 1, f^{(k)} is continuous on [a,b] (since f^{(k+1)} exists) and therefore belongs to \mathcal{R}[a,b]. Repeating this integration by parts n times yields
R_{k} = -\frac{f^{(k)}(a)}{k!}(b-a)^k + R_{k-1} \quad \text{for } k = n, n-1, \dots, 1,
down to R_0 = \int_a^b f'(t)\,dt = f(b) - f(a) by the First Form of the Fundamental Theorem 7.3.1.Induction backwards from k=n to k=1, ending with FTC 1 (T20) for R_0 = \int_a^b f'(t)\,dt. Rearranging R_{k-1} = \frac{f^{(k)}(a)}{k!}(b-a)^k + R_k from k=1 to k=n gives
f(b) - f(a) = R_0 = \frac{f'(a)}{1!}(b-a) + R_1 = \cdots = \sum_{k=1}^n \frac{f^{(k)}(a)}{k!}(b-a)^k + R_n,
which is Taylor's formula (8). ∎Back-substitute R_1, R_2, \dots, R_n into f(b) = f(a) + R_0.   Check yourself · Verify Theorem 7.3.18 for f(x) = e^x, a = 0, b = 1, n = 1 (Q36) Compute the Taylor polynomial f(0) + f'(0)(1-0) and the integral remainder R_1 = \int_0^1 e^t(1-t)\,dt, and check their sum. f(0) + f'(0)(1) = 1 + 1 = 2. By parts with F(t)=e^t, G(t)=1-t: R_1 = \big[e^t(1-t)\big]_0^1 - \int_0^1 e^t(-1)\,dt = (0 - 1) + (e - 1) = e - 2. Sum: 2 + (e - 2) = e = f(1). In the model paper: Q36 (PYQ MTS-25 Q19, MAT-22 Q32). </section>
<!-- COUNTEREXAMPLE LIBRARY PAGE --> <section class="page"> REFERENCE · COUNTEREXAMPLES <h1>Module 3 Counterexample Library</h1> Know these six C-coded functions and four auxiliary non-examples cold: every "Is the converse true?" or "Can this hypothesis be dropped?" question uses one of them. <table class="t"> <tr><th>Code & Function on [a,b]</th><th>Properties</th><th>What it disproves</th></tr> <tr><td>C7 Dirichlet on [0,1]: f(x)=1 (x\in\mathbb{Q}), 0 (x\notin\mathbb{Q})</td><td>Bounded (0\le f\le 1); takes only 2 values; discontinuous everywhere (D=[0,1] not null); S=1 (rational tags) vs S=0 (irrational tags).</td><td>Bounded \not\Rightarrow f\in\mathcal{R}[0,1] (T08); finitely many values \not\Rightarrow step function (T14); Cauchy & Lebesgue fail (T11, T27).</td></tr> <tr><td>C8 Thomae (popcorn) on [0,1]: h(m/n)=1/n (lowest terms), h(0)=1, h(x)=0 (x\notin\mathbb{Q})</td><td>h\in\mathcal{R}[0,1] with \int_0^1 h = 0; continuous at irrationals, discontinuous on D=\mathbb{Q}\cap[0,1] (null); H(x)=\int_0^x h \equiv 0.</td><td>f\in\mathcal{R} \not\Rightarrow continuous, uniformly continuous, or monotone (T15, T16); H'(x)=0\ne h(x) on \mathbb{Q}, so H is not an antiderivative of h (T24).</td></tr> <tr><td>C9 Signum / Step on [-1,1]: f(x)=\operatorname{sgn} x (-1,0,1)</td><td>Step function, monotone increasing, f\in\mathcal{R}[-1,1], \int_{-1}^1 \operatorname{sgn} = 0; discontinuous at 0.</td><td>f\in\mathcal{R} \not\Rightarrow continuous (T15); MVT for integrals fails without continuity (on [0,2], g=1 on [0,1), 2 on [1,2] never takes average value 3/2).</td></tr> <tr><td>C16 Unbounded 1/x on [0,1]: f(x)=1/x (x\in(0,1]), f(0)=0</td><td>Continuous on (0,1]; D=\{0\} is null; f|_{[c,1]}\in\mathcal{R}[c,1] for every c\in(0,1);

unbounded.</td><td>f\notin\mathcal{R}[0,1] by 7.1.6 (T08); continuous on (a,b] \not\Rightarrow \mathcal{R}[a,b] (T15); \mathcal{R}[c,b]\ \forall c>a \not\Rightarrow \mathcal{R}[a,b] without boundedness (T18); Lebesgue needs bounded (T27).</td></tr> <tr><td>C19 Unbounded derivative on [0,1]: K(x)=x^2\cos(1/x^2), K(0)=0</td><td>K is continuous and differentiable at every x\in[0,1] (E=\varnothing), but K'(x)=2x\cos(1/x^2)+(2/x)\sin(1/x^2) is unbounded.</td><td>K'\notin\mathcal{R}[0,1]: FTC 1 hypothesis (c) f\in\mathcal{R}[a,b] is NOT automatic even when F'=f everywhere (T20, T21).</td></tr> <tr><td>C20 Indefinite integral of \operatorname{sgn}: f=\operatorname{sgn} on [-1,1], F(x)=\lvert x\rvert - 1</td><td>F(x)=\int_{-1}^x \operatorname{sgn} = \lvert x\rvert - 1 is Lipschitz (M=1) and continuous, but F'(0) DNE (slopes \mp 1).</td><td>FTC 2 needs continuity of f at c: when f jumps at 0, F has a corner at 0 and is not an antiderivative on [-1,1] (T23, T24).</td></tr> <tr><td>Aux 1 · Countable spikes: G(1/n)=1/n vs H(1/k)=k (0 else on [0,1])</td><td>Both have D=\{1/k : k\in\mathbb{N}\} (countable, hence null). G is bounded (\le 1); H is unbounded.</td><td>G\in\mathcal{R}[0,1] with \int_0^1 G = 0 (7.1.4d, T06, T27); H\notin\mathcal{R}[0,1] (7.1.6, T08). Changing g=0 on a countable set can destroy integrability!</td></tr> <tr><td>Aux 2 · \pm 1 Dirichlet: f(x)=1 (x\in\mathbb{Q}), -1 (x\notin\mathbb{Q})</td><td>\lvert f(x)\rvert = 1 and f(x)^2 = 1 for all x\in[0,1], so \lvert f\rvert, f^2 \in \mathcal{R}[0,1], but f\notin\mathcal{R}[0,1].</td><td>Converse of Corollary 7.3.15 (\lvert f\rvert\in\mathcal{R} \not\Rightarrow f\in\mathcal{R}, T29) and converse of squaring (f^2\in\mathcal{R} \not\Rightarrow f\in\mathcal{R}, T30).</td></tr> <tr><td>Aux 3 · Bad outer \varphi: h Thomae, \varphi(0)=0, \varphi(y)=1 (y>0)</td><td>Both h \in \mathcal{R}[0,1] and \varphi \in \mathcal{R}[0,1], but \varphi is discontinuous at 0.</td><td>\varphi \circ h = Dirichlet \notin \mathcal{R}[0,1]: continuity of outer \varphi in Composition Theorem 7.3.14 cannot be dropped (T28).</td></tr> <tr><td>Aux 4 · Onepoint spike: f(0)=1, f(x)=0 on (0,1]</td><td>f(x) \ge 0 on [0,1] and \int_0^1 f = 0 (by 7.1.3), yet f(0) = 1 \ne 0.</td><td>Shows "f\ge 0 and \int_a^b f = 0 \Rightarrow f \equiv 0" fails if continuity of f is dropped (Bartle Ex 7.2.9, T15).</td></tr> </table> </section>
<!-- WHICH TOOL? MIXED DRILL --> <section class="page"> INTERLEAVED DRILL · WHICH TOOL? <h1>"Which tool?" mixed recognition drill</h1> In the exam, questions are not labelled by section. Read each prompt, name the exact theorem/tool and the hypotheses to check, then uncover the answer.
Drill 1 · Integrability of a messy bounded function Show that g(x) = \sin(1/x) (x \in (0,1]), g(0) = 0 is in \mathcal{R}[0,1]. Which two tools work? Tool A (Lebesgue's Criterion 7.3.12, T27): \lvert g(x)\rvert \le 1 (bounded) and g is discontinuous only at \{0\} (a 1-point null set). Tool B (Bartle Ex 7.2.11, T18): \lvert g\rvert \le 1 on [0,1] and g is continuous (hence integrable by 7.2.7) on [c,1] for every c \in (0,1).
Drill 2 · Disproving integrability Show that H(1/k) = k (k \in \mathbb{N}), 0 elsewhere on [0,1] is not in \mathcal{R}[0,1]; then show h(x) = x + 1 on \mathbb{Q}, 0 on irrationals is not in \mathcal{R}[0,1]. For H: Boundedness Theorem 7.1.6 (T08)—H is unbounded (H(1/k)=k\to\infty), so H\notin\mathcal{R}[0,1] in one line! For h: h is bounded (0\le h\le 2), so 7.1.6 does not apply; use the Cauchy Criterion negation (7.2.1, T11) with \varepsilon_0 = 1 (rational tags give S \ge 1, irrational tags give S = 0), or Lebesgue 7.3.12 (T27) (D = [0,1], not null).
Drill 3 · FTC 1 vs Substitution at a singular

endpoint Compare \int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt and \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt. Why do they need different theorems? On [1,4], \varphi(t) = \sqrt t has \varphi'(t) = \frac{1}{2\sqrt t} continuous on [1,4], so Substitution (Theorem 7.3.8, T25) works directly. On [0,4], \varphi'(0) does not exist, so 7.3.8 fails; instead use Ex 7.2.11 (T18) for integrability (\lvert\sin\sqrt t/\sqrt t\rvert \le 1) and FTC 1 (Theorem 7.3.1, T20) with F(t) = -2\cos\sqrt t and exceptional set E = \{0\}.
Drill 4 · Derivative of an integral with variable limits Find F'(x) for F(x) = \int_{x^2}^x \sqrt{1+t^2}\,dt. Which theorems do you cite? Split via Theorem 7.2.13 (T19) as F(x) = G(x) - G(x^2) where G(u) = \int_0^u \sqrt{1+t^2}\,dt. Cite FTC 2 (Theorem 7.3.6, T23) (since \sqrt{1+t^2} is continuous) to get G'(u) = \sqrt{1+u^2}, and the Chain Rule 6.1.6 (P11) on G(x^2) to get F'(x) = \sqrt{1+x^2} - 2x\sqrt{1+x^4}.
Drill 5 · Integral equals zero on all subintervals If f : [0,1] \to \mathbb{R} is continuous and \int_0^x f = \int_x^1 f for all x \in [0,1], how do you prove f \equiv 0? What if instead f \ge 0 is continuous and \int_a^b f = 0? First prompt: put F(x) = \int_0^x f; by Additivity 7.2.9 (T17), F(x) = F(1) - F(x) \Rightarrow F(x) = \tfrac12 F(1) constant, then differentiate via FTC 2 (7.3.6, T23) to get f(x) = F'(x) = 0. Second prompt: only one integral \int_a^b f = 0 is given, so use continuity at c + Monotonicity 7.1.5(c) + Lemma 7.2.4 (T15, Ex 7.2.8): if f(c) > 0, then f > \tfrac12 f(c) on some interval J of length \ell > 0, giving \int_a^b f \ge \tfrac12 f(c)\ell > 0. </section>
<!-- THEOREM SHEET (STATEMENTS AND HYPOTHESES ONLY) --> <section class="page"> REVISION · THEOREM SHEET <h1>All Definitions & Theorems (Word for Word)</h1> Cover the right column and recite each statement and its exact hypotheses before the exam. <table class="t"> <tr><th>Ref & Topic</th><th>Exact Statement & Hypotheses</th></tr> <tr><td>Def 7.1.1 T03 Riemann Integrable</td><td>f:[a,b]\to\mathbb{R} is in \mathcal{R}[a,b] with L=\int_a^b f iff \exists L\in\mathbb{R} such that \forall\varepsilon>0\ \exists\delta_\varepsilon>0: for every tagged partition \dot P of [a,b] with \lVert P\rVert<\delta_\varepsilon, \lvert S(f;\dot P)-L\rvert<\varepsilon.</td></tr> <tr><td>Thm 7.1.2 T04 Uniqueness</td><td>Hyp: f\in\mathcal{R}[a,b]. Concl: The value of \int_a^b f is uniquely determined.</td></tr> <tr><td>Thm 7.1.3 T06 Finite Change</td><td>Hyp: g\in\mathcal{R}[a,b] and f(x)=g(x) except for a finite number of points in [a,b]. Concl: f\in\mathcal{R}[a,b] and \int_a^b f = \int_a^b g.</td></tr> <tr><td>Thm 7.1.5 T07 Linearity & Order</td><td>Hyp: f,g\in\mathcal{R}[a,b], k\in\mathbb{R}. Concl: (a) kf\in\mathcal{R}[a,b], \int_a^b kf=k\int_a^b f; (b) f+g\in\mathcal{R}[a,b], \int_a^b(f+g)=\int_a^b f+\int_a^b g; (c) f(x)\le g(x) on [a,b] \Rightarrow \int_a^b f\le\int_a^b g.</td></tr> <tr><td>Thm 7.1.6 T08 Boundedness</td><td>Hyp: f\in\mathcal{R}[a,b]. Concl: f is bounded on [a,b].</td></tr> <tr><td>Thm 7.2.1 T10 Cauchy Criterion</td><td>f\in\mathcal{R}[a,b] \iff \forall\varepsilon>0\ \exists\eta_\varepsilon>0 such that for any tagged partitions \dot P,\dot Q of [a,b] with \lVert P\rVert<\eta_\varepsilon and \lVert Q\rVert<\eta_\varepsilon, \lvert S(f;\dot P)-S(f;\dot Q)\rvert<\varepsilon.</td></tr> <tr><td>Thm 7.2.3 T12 Squeeze

Theorem</td><td>f\in\mathcal{R}[a,b] \iff \forall\varepsilon>0\ \exists\alpha_\varepsilon,\omega_\varepsilon\in\mathcal{R}[a,b] with \alpha_\varepsilon(x)\le f(x)\le\omega_\varepsilon(x) on [a,b] and \int_a^b(\omega_\varepsilon-\alpha_\varepsilon)<\varepsilon.</td></tr> <tr><td>Lem 7.2.4 / Thm 7.2.5 T13T14</td><td>If J\subseteq[a,b] has endpoints c<d, then \varphi_J\in\mathcal{R}[a,b] and \int_a^b\varphi_J=d-c. Every step function \varphi=\sum_{j=1}^m k_j\varphi_{J_j} is in \mathcal{R}[a,b] with \int_a^b\varphi=\sum_{j=1}^m k_j(d_j-c_j).</td></tr> <tr><td>Thm 7.2.7 / 7.2.8 T15T16</td><td>7.2.7: f continuous on [a,b] \Rightarrow f\in\mathcal{R}[a,b]. 7.2.8: f monotone on [a,b] \Rightarrow f\in\mathcal{R}[a,b].</td></tr> <tr><td>Thm 7.2.9 T17 Additivity</td><td>Hyp: f:[a,b]\to\mathbb{R}, c\in(a,b). Concl: f\in\mathcal{R}[a,b] \iff f|_{[a,c]}\in\mathcal{R}[a,c] and f|_{[c,b]}\in\mathcal{R}[c,b], and then \int_a^b f = \int_a^c f + \int_c^b f. (7.2.10: [c,d]\subseteq[a,b] \Rightarrow f|_{[c,d]}\in\mathcal{R}[c,d].)</td></tr> <tr><td>Thm 7.3.1 T20 FTC First Form</td><td>Hyp: E\subset[a,b] finite, (a) F continuous on [a,b], (b) F'(x)=f(x) for all x\in[a,b]\setminus E, (c) f\in\mathcal{R}[a,b]. Concl: \int_a^b f = F(b)-F(a).</td></tr> <tr><td>Thm 7.3.4 / 7.3.5 / 7.3.6 T22T23 FTC Second Form</td><td>Let f\in\mathcal{R}[a,b] and F(z):=\int_a^z f. 7.3.4: \lvert f\rvert\le M \Rightarrow \lvert F(z)-F(w)\rvert\le M\lvert z-w\rvert (F continuous). 7.3.5: If f is continuous at c\in[a,b], then F'(c)=f(c). 7.3.6: If f is continuous on [a,b], then F'(x)=f(x) on [a,b].</td></tr> <tr><td>Thm 7.3.8 T25 Substitution</td><td>Hyp: \varphi:[\alpha,\beta]\to\mathbb{R} has continuous derivative \varphi' on [\alpha,\beta]; f:I\to\mathbb{R} is continuous on an interval I\supseteq\varphi([\alpha,\beta]). Concl: \int_\alpha^\beta f(\varphi(t))\varphi'(t)\,dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx.</td></tr> <tr><td>Def 7.3.10 / Thm 7.3.12 T26T27 Lebesgue Criterion</td><td>Z\subseteq\mathbb{R} is null if \forall\varepsilon>0\ \exists\{(a_k,b_k)\}_{k=1}^\infty with Z\subseteq\bigcup(a_k,b_k) and \sum(b_k-a_k)\le\varepsilon. 7.3.12: Bounded f:[a,b]\to\mathbb{R} is in \mathcal{R}[a,b] \iff f is continuous a.e. on [a,b] (D(f) is null).</td></tr> <tr><td>Thm 7.3.14 / 7.3.15 / 7.3.16 T28–T30</td><td>7.3.14: f\in\mathcal{R}[a,b], f([a,b])\subseteq[c,d], \varphi:[c,d]\to\mathbb{R} continuous \Rightarrow \varphi\circ f\in\mathcal{R}[a,b]. 7.3.15: f\in\mathcal{R}[a,b] \Rightarrow \lvert f\rvert\in\mathcal{R}[a,b] and \lvert\int_a^b f\rvert\le\int_a^b\lvert f\rvert\le M(b-a). 7.3.16: f,g\in\mathcal{R}[a,b] \Rightarrow fg\in\mathcal{R}[a,b].</td></tr> <tr><td>Thm 7.3.17 / 7.3.18 T31T32 Parts & Taylor</td><td>7.3.17: F,G differentiable on [a,b], f=F', g=G'\in\mathcal{R}[a,b] \Rightarrow \int_a^b fG = [FG]_a^b - \int_a^b Fg. 7.3.18: f',\dots,f^{(n+1)} exist on [a,b], f^{(n+1)}\in\mathcal{R}[a,b] \Rightarrow f(b)=\sum_{k=0}^n\frac{f^{(k)}(a)}{k!}(b-a)^k + R_n, R_n=\frac{1}{n!}\int_a^b f^{(n+1)}(t)(b-t)^n\,dt.</td></tr> </table> </section>

<!-- RECALL PLAN & PYQ TABLE --> <section class="page"> Spaced Recall Plan · Day 0 / 1 / 3 / 7 / 14 <ul> Day 0 (today): Read through with a blank sheet; cover the proof of 7.2.7 (T15), 7.2.8 (T16), 7.2.9 (T17), and 7.3.5 (T23) and re-explain every "Why" line aloud. Day 1: Write from memory the exact definitions of \mathcal{R}[a,b] (7.1.1), Cauchy Criterion (7.2.1), Squeeze Theorem

(7.2.3), Null Set (7.3.10), and Lebesgue's Criterion (7.3.12). Day 3: Write out the four 10-mark proofs (Q37–Q40: Continuous \Rightarrow\mathcal{R}, Monotone \Rightarrow\mathcal{R}, Additivity, FTC 2) from their skeletons without looking. Day 7: Do the "Which tool?" drill above and Section A (Q1–Q20) of the Model Paper. Day 14 / Exam week: Recite the Counterexample Library (C7–C20) and sit Sections B and C of the Model Paper timed.
<h2>Past Questions (PYQ) → Topic & Model Paper Map</h2> All 74 in-scope Module 3 questions from MTS 6B 10 (2022, 2025, 2026) and MAT 6B 09 (2021, 2022). <table class="t"> <tr><th>Past Paper Tags (74 total)</th><th>Topic</th><th>Paper Q</th></tr> <tr><td>MTS-26 Q4; MAT-22 Q3, MCQ 16; MAT-21 Q5</td><td>T01 Partition & norm</td><td>Q1</td></tr> <tr><td>MTS-22 Q4</td><td>T02 Riemann sum</td><td>Q2</td></tr> <tr><td>MAT-21 MCQ 9</td><td>T03 Def 7.1.1 (\mathcal{R}[a,b])</td><td>Q3</td></tr> <tr><td>MTS-25 Q4</td><td>T05 Constant function \in\mathcal{R}</td><td>Q5</td></tr> <tr><td>MAT-22 MCQ 18; MAT-21 MCQ 13; MTS-22 MCQ 7</td><td>T06 Finite exception set (7.1.3)</td><td>Q6</td></tr> <tr><td>MTS-25 Q5; MAT-21 Q29; MTS-22 Q5; MAT-22 MCQ 14</td><td>T07 Linearity & monotonicity (7.1.5)</td><td>Q21</td></tr> <tr><td>MAT-22 MCQ 15; MAT-21 MCQ 11, 17; MTS-22 MCQ 8, 9</td><td>T08 Boundedness Theorem (7.1.6)</td><td>Q7</td></tr> <tr><td>MAT-21 Q8</td><td>T10 Cauchy Criterion (7.2.1)</td><td>Q23</td></tr> <tr><td>MAT-22 Q4, MCQ 17; MAT-21 MCQ 12, 16</td><td>T11 Dirichlet & x+1 not in \mathcal{R}</td><td>Q24</td></tr> <tr><td>MTS-26 Q18; MTS-22 Q6; MAT-21 Q7</td><td>T12 Squeeze Theorem (7.2.3)</td><td>Q25</td></tr> <tr><td>MAT-21 MCQ 10</td><td>T13 Elementary step \varphi_J (7.2.4)</td><td>Q8</td></tr> <tr><td>MTS-26 Q5; MAT-22 Q16</td><td>T14 Step functions \in\mathcal{R} (7.2.5)</td><td>Q26</td></tr> <tr><td>MTS-25 Q18; MAT-21 Q30; MTS-22 Q19; MAT-22 MCQ 21, 23; MAT-21 MCQ 15</td><td>T15 Continuous \Rightarrow\mathcal{R} (7.2.7) & \int f=0</td><td>Q18 Q37</td></tr> <tr><td>MTS-26 Q19; MAT-22 Q30, MCQ 22</td><td>T16 Monotone \Rightarrow\mathcal{R} (7.2.8)</td><td>Q38</td></tr> <tr><td>MTS-26 Q25; MAT-22 Q37a; MTS-22 Q25</td><td>T17 Additivity Theorem (7.2.9)</td><td>Q39</td></tr> <tr><td>MAT-22 Q37b; MAT-22 MCQ 24, 25; MAT-21 MCQ 18</td><td>T18 Restriction (7.2.10) & [c,b] (Ex 7.2.11)</td><td>Q27 Q28</td></tr> <tr><td>MTS-26 Q8; MAT-22 Q35, MCQ 27, 28, 29; MAT-21 MCQ 19, 20</td><td>T20 T21 FTC First Form (7.3.1) & evaluations</td><td>Q14 Q15 Q29</td></tr> <tr><td>MAT-21 Q6</td><td>T22 Indefinite integral (7.3.3, 7.3.4)</td><td>Q11</td></tr>

<tr><td>MTS-25 Q25; MAT-21 Q37a; MAT-22 Q33, MCQ 30; MTS-22 Q14</td><td>T23 FTC Second Form (7.3.5) & Leibniz</td><td>Q17 Q19 Q40</td></tr> <tr><td>MTS-25 Q6; MAT-22 Q18; MAT-21 Q19; MAT-22 MCQ 19, 26</td><td>T25 Substitution Theorem (7.3.8)</td><td>Q16 Q30</td></tr> <tr><td>MTS-25 Q7; MTS-26 Q7; MTS-22 MCQ 10</td><td>T26 Null sets & \mathbb{Q}\cap[0,1] (7.3.10, 7.3.11)</td><td>Q13</td></tr> <tr><td>MAT-22 Q6</td><td>T27 Lebesgue Criterion (7.3.12)</td><td>Q31</td></tr> <tr><td>MAT-21 Q4</td><td>T28 Composition Theorem (7.3.14)</td><td>Q32</td></tr> <tr><td>MTS-25 Q8; MTS-22 Q7; MAT-21 Q20</td><td>T29 Absolute Value \lvert f\rvert\in\mathcal{R} (7.3.15)</td><td>Q33</td></tr> <tr><td>MTS-26 Q6; MAT-22 Q17; MAT-21 Q18</td><td>T30 Product Theorem fg\in\mathcal{R} (7.3.16)</td><td>Q34</td></tr> <tr><td>MTS-25 Q19; MAT-22 Q32</td><td>T32 Taylor's Theorem (7.3.18)</td><td>Q36</td></tr> </table> </section>
=== NOTES / DOUBTS (STAGE 3 OF 6) ===

- Completed Stage 3 (src/m3/prepare_04_sec73.html): covers Unit 3 (Bartle §7.3, topics T20–T32 corresponding to K20–K32), plus the Counterexample Library page, the "Which tool?" mixed drill, the word-for-word Theorem Sheet, the Spaced Recall Plan (Day 0/1/3/7/14), and the PYQ → Topic table covering all 74 past questions.
- Includes all known issues from the prompt: Bartle 7.3.18 proof misprint (b-a)^{n-1} \to (b-t)^{n-1} noted and corrected; Bartle 7.3.9(b) \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt = F(4)-F(0) = 2(1-\cos 2) = 2.832294 via FTC 1 with E=\{0\}; all numerical values from brief §0 and §4 verified.
- Note on PYQ count: content_m3.md §5 lists 74 tags across 42 rows, while pyq_all.md Part 8 has 3 extra MTS6B10-22 rows at the bottom (MTS-22 Q14 Leibniz rule, MTS-22 Q19 continuous \Rightarrow \mathcal{R}, MTS-22 Q25 Additivity theorem), making 77 rows in Part 8. Both the 74-tag list from content_m3.md §5 and the 3 additional MTS-22 tags (MTS-22 Q14, Q19, Q25) are included in the PYQ map above and will appear in src/m3/paper.html in Stage 4!
- Next up in Stage 4: src/m3/paper.html (the 40-question model paper + coverage table).
- Reminder for Stage 6: Please paste or attach figs.py and figs_m2.py before or when requesting Stage 6 so src/m3/figs_m3.py can be generated using the exact S and Plot classes.